Dyadic Data Analysis with R

Part II: Applied Tutorial

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Today’s aims and program

Time	Focus
11:00–12:45	Introduction and types of dyadic models
12:45–14:00	*Lunch break*
14:00–15:30	Estimating cross-sectional dyadic models in R
15:30–15:45	*Short break*
15:45–16:45	Intensive longitudinal extensions
16:45–17:00	*Short break*
17:00–18:00	Exercises in R

Software and frameworks

SEM vs. MLM

- Many simple dyadic models can be represented in either framework.
- The best framework is the one that represents your conceptual model most naturally (Ledermann & Kenny, 2017).

SEM / DSEM is often more natural for

- Latent dyadic constructs (e.g., a CFM)
- Item-level measurement models for reflective measures
- Linked outcomes or equations [e.g., mediation; Ledermann & Macho (2009)]
- Short dynamic panels in wide-format [e.g., LD-APIM; Gistelinck et al. (2021)]
- Intensive lagged or coupled dynamics via DSEM/RDSEM (e.g., Asparouhov et al., 2018)

MLM is often more natural for

- Parsimonious observed-variable models with modest numbers of dyads
- Nested grouping and flexible random-effect structures
- Observed-outcome growth or concurrent ILD models with flexible time trends
- Complex observed-variable regressions—interactions, polynomials, splines, and dyadic RSA—especially with repeated measures or additional nesting
- Specialized generalized mixed models in R, including zero-inflated, hurdle, and Tweedie models

(Asparouhov et al., 2018; Asparouhov & Muthén, 2020; Brooks et al., 2017; del Rosario & West, 2025; Driver et al., 2017; Gistelinck et al., 2021; Nestler et al., 2019; Schönbrodt et al., 2018)

SEM vs. MLM: Missing data

- **MLM: Complete-case**
  - **Outcome missing:** Valid under MAR, MLM uses remaining observations
  - **Predictor missing:** Valid under MAR, if missingness does **not depend** on the outcome after accounting for predictors
- **SEM with FIML and joint likelihood**
  - Valid under MAR: Models predictors and outcomes jointly
- **MI: Imputation**
  - Handles missing outcomes, predictors, and auxiliary information
  - Must match the analysis and preserve person, dyad, role, and time structure

Include important predictors of missingness and missing values. FIML and MI cover more MAR situations than row omission; none automatically solves MNAR.

Ledermann & Kenny (2017); Hughes et al. (2019); Grund et al. (2018)

Why R

Why we teach R

- **Free to use:** no license fees or institutional access barriers
- **Open source:** statistical implementations can be inspected, audited, and adapted
- **Reproducible:** workflows can be rerun, reviewed, shared, and version-controlled
- **Extensible:** specialized methods integrate with data, graphics, and reporting

Current trade-offs

- **SEM/DSEM:** R spreads these capabilities across specialized packages
  - `lavaan`: fixed-wave and limited two-level SEM
  - `mlts` and `ctsem`: multivariate dynamic models
    - Dyadic DSEM in R remains cumbersome and limited
    - Mplus remains the most complete (non-open) option with DSEM capabilities
- **MLM/ILD:** `glmmTMB` and `brms` both model serial dependence
  - Neither directly offers a separable partner-by-time residual structure such as $\mathrm{UN} \otimes \mathrm{AR}(1)$
    - SAS `PROC MIXED` and SPSS `MIXED` do for Gaussian repeated-measures models

(Asparouhov et al., 2018; Asparouhov & Muthén, 2020; Bürkner, n.d.; Driver et al., 2017; IBM Corp., n.d.; Koslowski et al., 2025; Kristensen & McGillycuddy, 2023; lavaan project, n.d.; Merkle et al., n.d.; Mulder & Hamaker, 2021; SAS Institute Inc., n.d.)

Estimating cross-sectional dyadic models in R

One possible workflow

1. **Define dyads, distinguishability, and research questions**
2. **Prepare the data and validate the dyadic structure**
3. **Describe and visualize measures and observed relationships**
4. **Estimate the models**
5. **Verify model adequacy, distinguishability (if applicable), and robustness**
6. **Visualize, interpret, and report results**

Synthesized from Kenny et al. (2006), Ledermann & Kenny (2015), and Stas et al. (2018).

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The Dataset

Same as before:

- 36 physically inactive romantic couples
- Both partners intended to become more physically active
- Smartphone-based planning interventions and JITAIs
- Daily diaries over 55 days

Aggregated for cross-sectional analyses

Research Question

RQ: In romantic couples, is a person's self-efficacy linked to physical activity in both partners?

Measurement - predictor

Self-efficacy

> **Rough English translation of the original German item**
> "I am confident that I can be physically active tomorrow, even if it becomes difficult."
>
> Adapted from Sniehotta et al. (2005)

0 = *not at all true today*

5 = *completely true today*

Measurement - outcome

Moderate to vigorous physical activity (MVPA)

> **Rough English translation of the original German item**
> “How many minutes did you spend engaging in moderate-to-vigorous physical activity today?”
>
> Adapted from Amireault & Godin (2015)

- **Alone:** _____ minutes
- **Together with your partner:** _____ minutes

Total MVPA = solo MVPA + joint MVPA.

Distinguishable dyads

Distinguishing by gender

Workflow

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Load and inspect the raw data

```r
library(dplyr) # for easier data-wrangling, exports functions like `select()`

raw_dyad_data <- readRDS("dyadic-person-means.rds") |>
  dplyr::select(
    couple_id, person_id, gender, efficacy, total_mvpa
  )

# usually use head(raw_dyad_data), slide_table is a custom function for html tables on these slides
slide_table(raw_dyad_data[1:4,])
```

couple_id	person_id	gender	efficacy	total_mvpa
1.00	1.00	male	3.02	33.61
1.00	2.00	female	0.89	12.71
2.00	3.00	male	0.13	42.83
2.00	4.00	female	2.37	44.54

This dataset is stored as an R object. Other packages provide functions to import various formats, such as:
`readxl::read_excel()`, `utils::read.csv()`, `haven::read_sav()`, `haven::read_sas()`

Create the required format and validate the data

For the distinguishable APIM we need to add:

- `efficacy_actor` and `efficacy_partner` columns
- Dummy coded numeric variables for the two roles

Validate things like:

- no dyad with more than two members,
- each dyad-member combination occurs only once (per occasion),
- gender labels are present and consistent
- you only have the type of dyad you expect

Create the required format and validate the data

We can use the `dyadMLM` package to help with data preparation and validation.

```r
library(dyadMLM)

apim_distinguishable_data <- dyadMLM::prepare_dyad_data( # exported function from dyadMLM
  data = raw_dyad_data,
  dyad = couple_id,
  member = person_id,
  role = gender,
  predictors = efficacy,
  model_types = 'apim',
  # if we want to fit an exchangeable model from the same data we can:
  include_arbitrary_member_contrast = TRUE,
  add_apim_gmc_predictors = TRUE,
  seed = 123
)

print(apim_distinguishable_data)
```

Create the required format and validate the data

```
# dyadMLM data
# Rows: 72 | Dyads: 36 | Intensive longitudinal: no
# Structure: dyad = couple_id, member = person_id, role = gender
#
# Dyad compositions:
# female_x_male distinguishable 36 dyads
#
# Added columns:
#   .composition                 inferred dyad composition
#   .composition_role            composition-specific member role
#   .is_{role}                   composition-role indicator columns
#   .member_contrast_arbitrary   composition-specific member contrasts coded
#                                -1/+1 in arbitrary direction for
#                                exchangeability-constrained random effects.
#                                Values are 0 for other compositions
#   .{pred}_actor                APIM actor predictor: actor's original
#                                predictor values
#   .{pred}_partner              APIM partner predictor: partner's original
#                                predictor values
#   .{pred}_gmc                  APIM grand-mean-centered predictor source:
#                                original values minus the mean across all
#                                retained non-missing observations
#   .{pred}_gmc_actor            APIM grand-mean-centered actor predictor:
```

Create the required format and validate the data

Manual code →

Looking at the columns immediately relevant to us:

```r
head(apim_distinguishable_data) |>
  dplyr::select(
    couple_id, person_id, gender, .is_female, .is_male, .efficacy_gmc, .efficacy_gmc_actor, .efficacy
  ) |>
  slide_table()
```

couple_id	person_id	gender	.is_female	.is_male	.efficacy_gmc	.efficacy_gmc_actor	.efficacy_gmc_partner	tota
1.00	1.00	male	0.00	1.00	0.64	0.64	-1.49	
1.00	2.00	female	1.00	0.00	-1.49	-1.49	0.64	
2.00	3.00	male	0.00	1.00	-2.24	-2.24	-0.01	
2.00	4.00	female	1.00	0.00	-0.01	-0.01	-2.24	
3.00	5.00	female	1.00	0.00	0.96	0.96	-0.76	
3.00	6.00	male	0.00	1.00	-0.76	-0.76	0.96	

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Create a wide view for role-specific exploration

The models will **NOT** use this dataset

```r
apim_distinguishable_wide_data <- apim_distinguishable_data |>
  dplyr::select(couple_id, gender, efficacy, total_mvpa) |>
  tidyr::pivot_wider(
    names_from = gender,
    values_from = c(efficacy, total_mvpa)
  )

slide_table(head(apim_distinguishable_wide_data))
```

couple_id	efficacy_male	efficacy_female	total_mvpa_male	total_mvpa_female
1.00	3.02	0.89	33.61	12.71
2.00	0.13	2.37	42.83	44.54
3.00	1.62	3.33	32.29	62.86
4.00	1.09	0.75	21.55	51.89
5.00	1.35	1.48	11.35	9.79
6.00	1.18	0.63	7.55	3.43

Describe focal variables per role

```r
library(report)

apim_distinguishable_wide_data |>
  dplyr::select(-couple_id) |>
  report::report_table() |>
  slide_table()
```

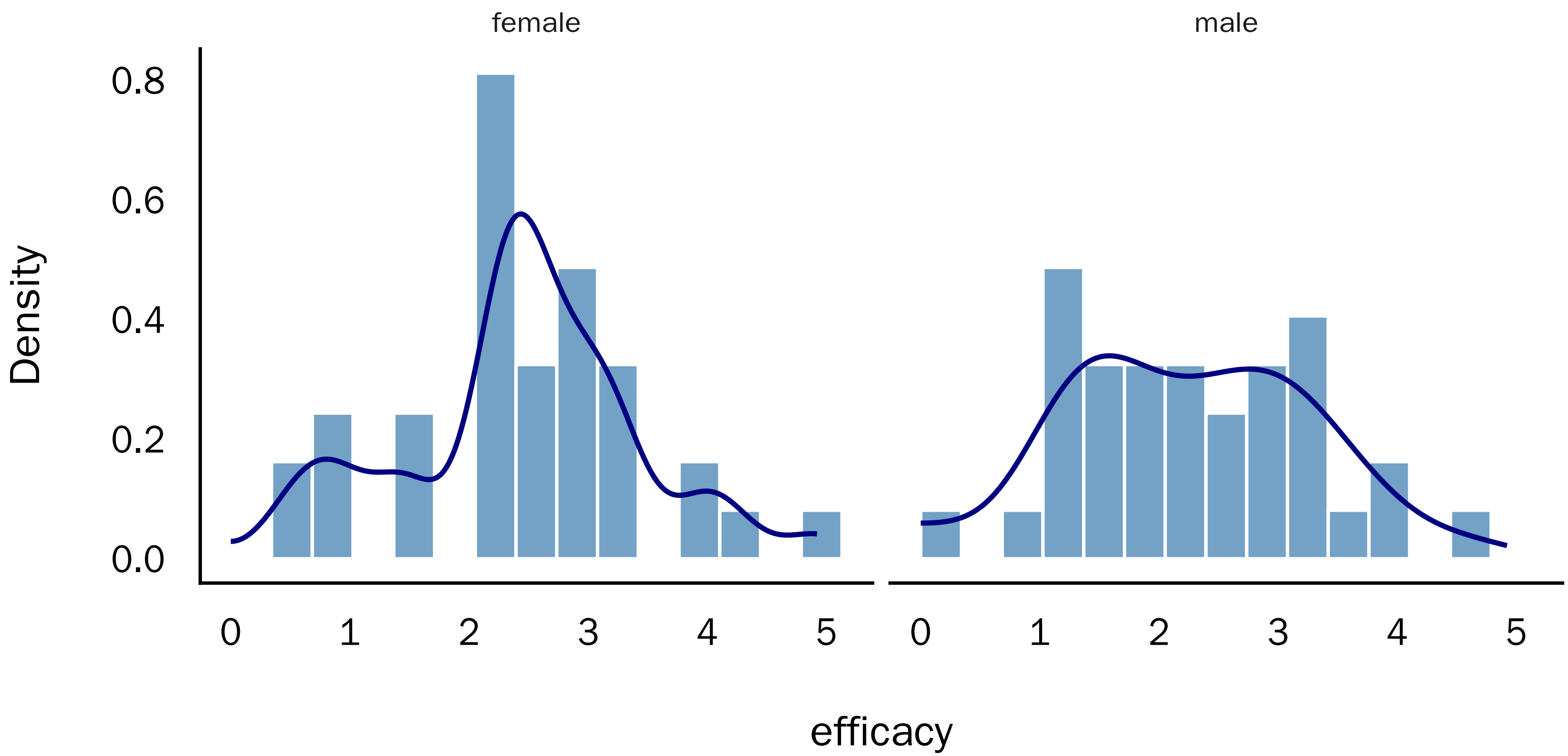
Variable	n_Obs	Mean	SD	Median	MAD	Min	Max	Skewness	Kurtosis	n_Missing
efficacy_male	36.00	2.29	0.99	2.32	1.18	0.13	4.60	0.16	-0.39	0.00
efficacy_female	36.00	2.46	1.01	2.39	0.85	0.46	4.92	0.01	0.28	0.00
total_mvpa_male	36.00	28.84	20.28	27.26	16.82	3.51	111.16	2.11	7.00	0.00
total_mvpa_female	36.00	33.58	24.39	29.85	20.90	3.43	126.35	1.74	4.80	0.00

Visualize focal variable by partner role

```
library(ggplot2)
library(see)

plot_role_histograms <- function(data, group, var) {
  var_label <- rlang::as_label(rlang::enquo(var))

  ggplot(
    data,
    aes(x = {{ var }}, y = after_stat(density))
  ) +
    geom_histogram(
      bins = 15, boundary = 0,
      fill = "steelblue", color = "white", alpha = 0.75
    ) +
    geom_density(
      bounds = c(0, Inf), color = "navy", linewidth = 1.1
    ) +
    facet_wrap(vars({{ group }}), nrow = 1) +
    labs(x = var_label, y = "Density") +
    see::theme_modern(base_size = 16)
}
```

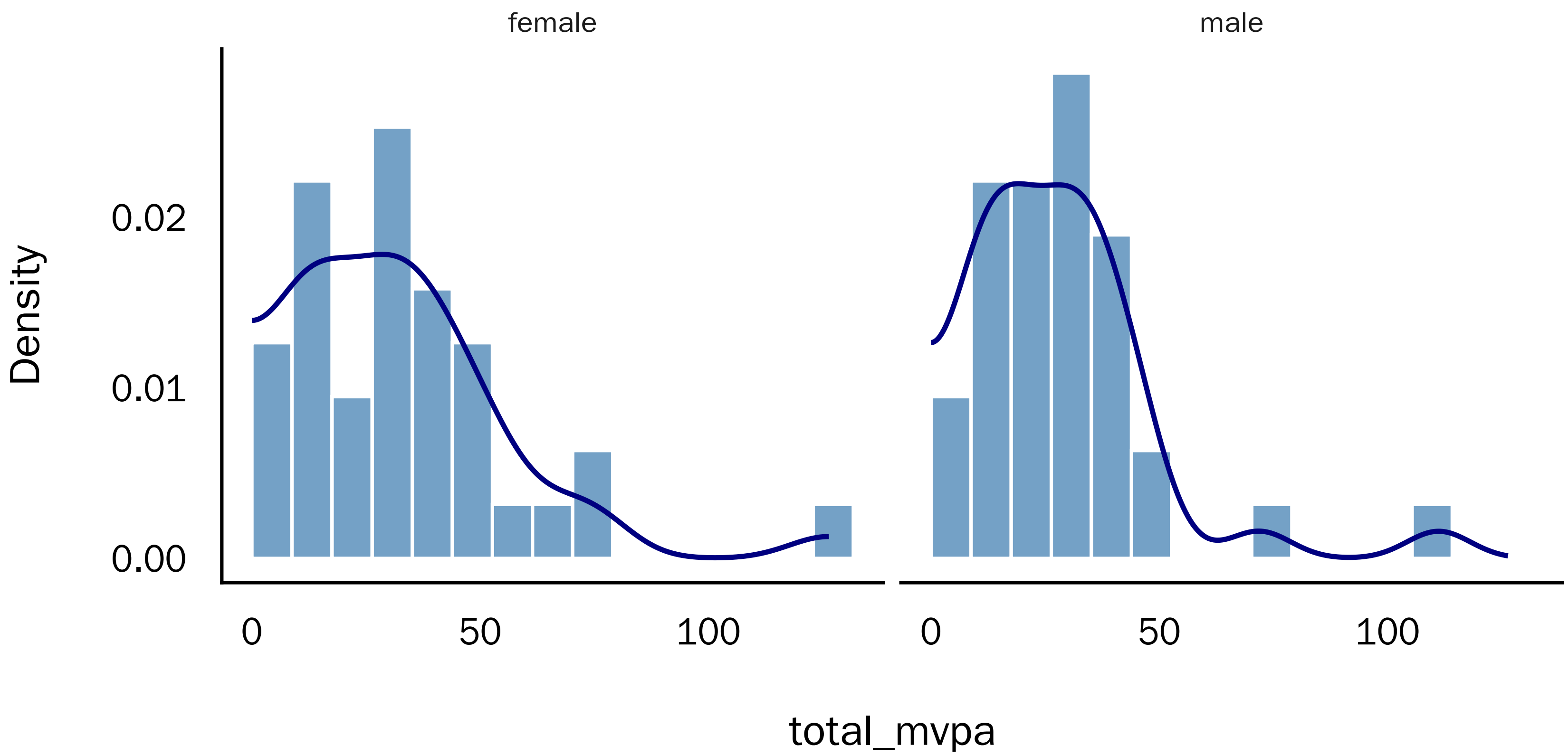
Visualize focal variable by partner role - Efficacy

```r
plot_role_histograms(apim_distinguishable_data, gender, efficacy)
```



Visualize focal variable by partner role - MVPA

```
plot_role_histograms(apim_distinguishable_data, gender, total_mvpa)
```



Observed dyadic correlations

The wide data ensure that each couple contributes once.

```r
library(correlation)
library(insight)

apim_distinguishable_wide_data |>
  correlation::correlation(
    select = c("efficacy_female", "total_mvpa_female"),
    select2 = c("efficacy_male", "total_mvpa_male")
  ) |>
  summary() |>
  insight::print_md(footer = "")
```

Correlation Matrix (pearson-method)

Parameter	efficacy_male	total_mvpa_male
efficacy_female	0.60***	0.43*
total_mvpa_female	0.42*	0.82***

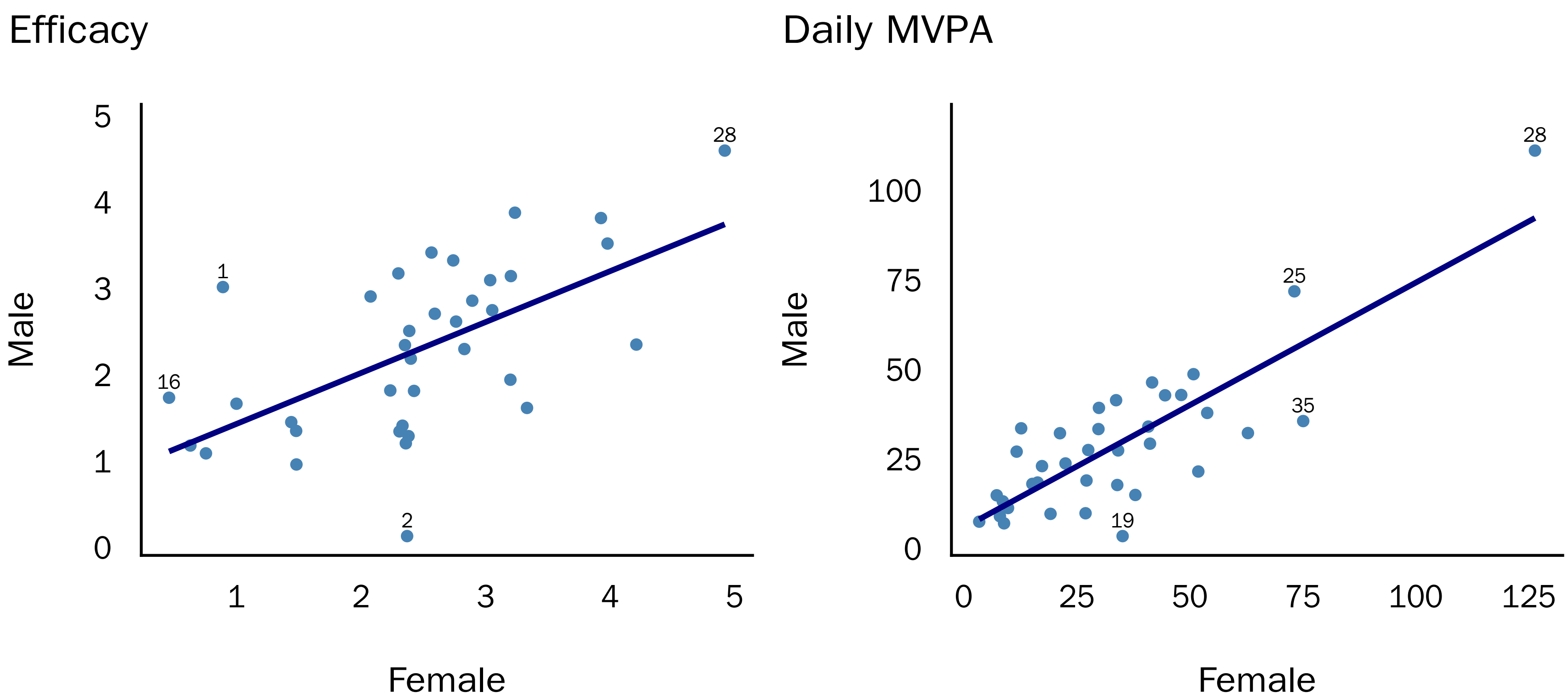
Visualize partner similarity

```r
plot_partner_similarity <- function(data, x, y, id, title, n_mahalanobis = 3) {
  x_name <- rlang::as_name(rlang::enquo(x))
  y_name <- rlang::as_name(rlang::enquo(y))
  xy <- cbind(data[[x_name]], data[[y_name]])
  labels <- data |>
    dplyr::mutate(
      .distance = mahalanobis(xy, colMeans(xy), cov(xy))
    ) |>
    dplyr::slice_max(.distance, n = n_mahalanobis)

  ggplot(data, aes(x = {{ x }}, y = {{ y }})) +
    geom_point(color = "steelblue") +
    geom_smooth(method = "lm", se = FALSE, color = "navy") +
    geom_text(
      data = labels,
      aes(label = {{ id }}),
      vjust = -0.6, check_overlap = TRUE, size = 4
    ) +
    labs(title = title, x = "Female", y = "Male") +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.12))) +
    see::theme_modern(base_size = 16)
}
```

Visualize partner similarity

```r
n_mahalanobis <- 4

see::plots(
  plot_partner_similarity(
    apim_distinguishable_wide_data,
    efficacy_female, efficacy_male, couple_id, "Efficacy",
    n_mahalanobis = n_mahalanobis
  ),
  plot_partner_similarity(
    apim_distinguishable_wide_data,
    total_mvpa_female, total_mvpa_male, couple_id, "Daily MVPA",
    n_mahalanobis = n_mahalanobis
  )
)
```

Visualize partner similarity



Labels: four largest Mahalanobis distances in each plot.

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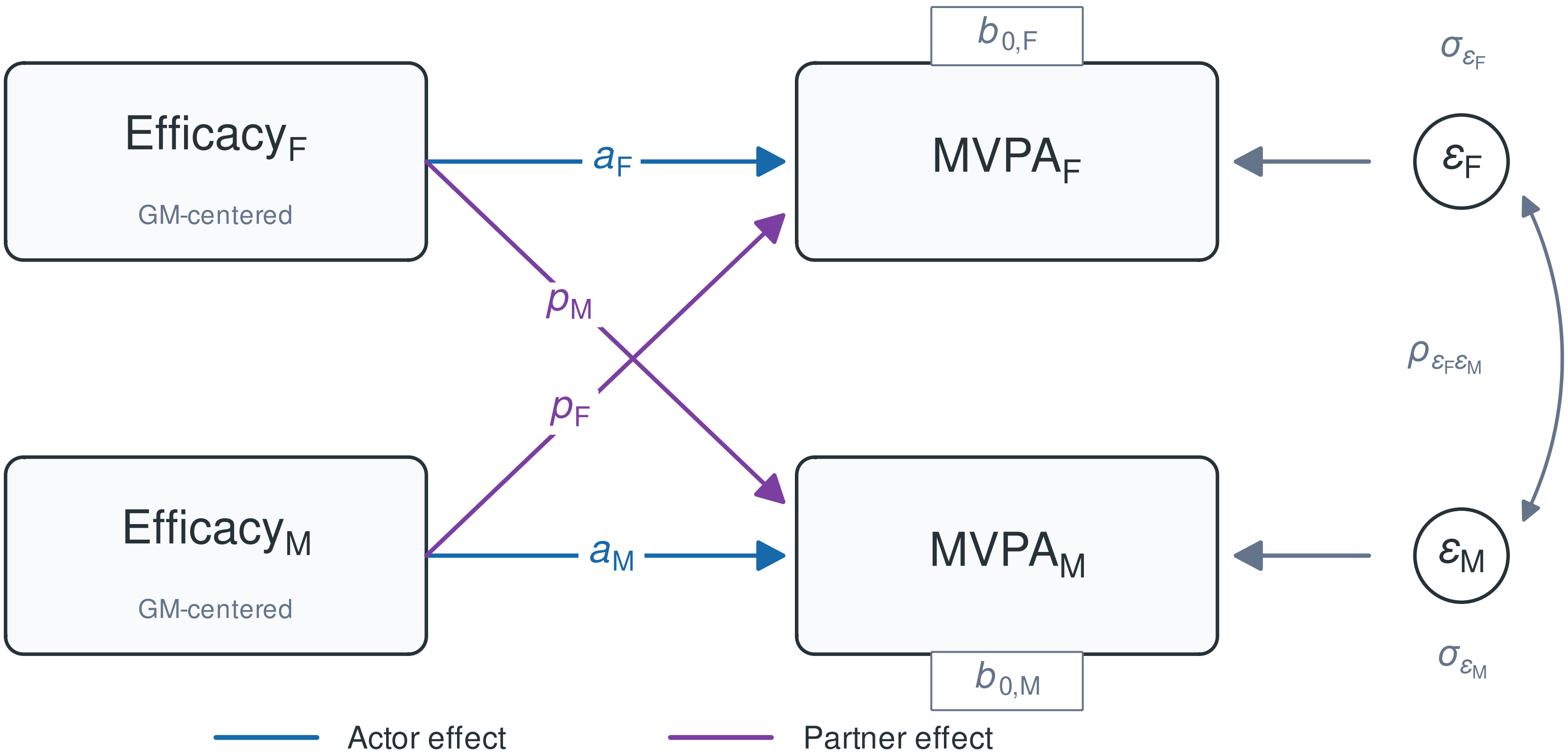
Distinguishable APIM



The APIM starts with two member-specific equations

For each dyad i, each member with role $r \in \{\mathrm{M}, \mathrm{F}\}$ has their own predictor $X_{ri}^{(\mathrm{A})}$ and their partner's predictor $X_{ri}^{(\mathrm{P})}$:

$$Y_{\mathrm{M},i} = b_{0,\mathrm{M}} + a_{\mathrm{M}} \times X_{\mathrm{M},i}^{(\mathrm{A})} + p_{\mathrm{M}} \times X_{\mathrm{M},i}^{(\mathrm{P})} + \epsilon_{\mathrm{M},i}$$
$$Y_{\mathrm{F},i} = b_{0,\mathrm{F}} + a_{\mathrm{F}} \times X_{\mathrm{F},i}^{(\mathrm{A})} + p_{\mathrm{F}} \times X_{\mathrm{F},i}^{(\mathrm{P})} + \epsilon_{\mathrm{F},i}$$

- Actor effects - Partner effects

In long format, each equation applies to one member row of the dyad.

How can we combine these two equations into one model?

Role indicators combine the equations

The indicators $D_{\mathrm{M},ri}$ and $D_{\mathrm{F},ri}$ identify the male or female row:

$$Y_{ri} = \underbrace{b_{0,\mathrm{M}} \times D_{\mathrm{M},ri} + b_{0,\mathrm{F}} \times D_{\mathrm{F},ri}}_{\text{role-specific intercepts}}$$

$$+ \underbrace{a_{\mathrm{M}} \times D_{\mathrm{M},ri} \times X_{ri}^{(\mathrm{A})} + a_{\mathrm{F}} \times D_{\mathrm{F},ri} \times X_{ri}^{(\mathrm{A})}}_{\text{role-specific actor effects}}$$

$$+ \underbrace{p_{\mathrm{M}} \times D_{\mathrm{M},ri} \times X_{ri}^{(\mathrm{P})} + p_{\mathrm{F}} \times D_{\mathrm{F},ri} \times X_{ri}^{(\mathrm{P})}}_{\text{role-specific partner effects}} + \epsilon_{ri}.$$

For a male row: $D_{\mathrm{M}} = 1,\ D_{\mathrm{F}} = 0$. For a female row: the reverse.

Residuals remain correlated within dyads

The two member rows from the same dyad retain a joint residual structure:

$$\mathrm{Cov}\begin{pmatrix}\epsilon_{\mathrm{F}i}\\ \epsilon_{\mathrm{M}i}\end{pmatrix} = \boldsymbol{\Sigma}_{\epsilon} = \begin{bmatrix}\sigma^2_{\epsilon_{\mathrm{F}}} & \rho_{\epsilon_{\mathrm{F}}\epsilon_{\mathrm{M}}}\sigma_{\epsilon_{\mathrm{F}}}\sigma_{\epsilon_{\mathrm{M}}}\\ \rho_{\epsilon_{\mathrm{F}}\epsilon_{\mathrm{M}}}\sigma_{\epsilon_{\mathrm{F}}}\sigma_{\epsilon_{\mathrm{M}}} & \sigma^2_{\epsilon_{\mathrm{M}}}\end{bmatrix}$$

Fitting the distinguishable APIM in glmmTMB

```r
library(glmmTMB)

apim_distinguishable_model <- glmmTMB(
  total_mvpa ~
    # Role-specific intercepts
    0 + .is_female + .is_male +

    # Role-specific actor effects
    .is_female:.efficacy_gmc_actor +
    .is_male:.efficacy_gmc_actor +

    # Role-specific partner effects
    .is_female:.efficacy_gmc_partner +
    .is_male:.efficacy_gmc_partner +

    # Within-dyad residual covariance
    (0 + .is_female + .is_male | couple_id),
  dispformula = ~ 0,
  family = gaussian(),
  data = apim_distinguishable_data
)

summary(apim_distinguishable_model)
```

Fitting the distinguishable APIM in glmmTMB

```
 Family: gaussian  ( identity )
Formula:
total_mvpa ~ 0 + .is_female + .is_male + .is_female:.efficacy_gmc_actor +
    .is_male:.efficacy_gmc_actor + .is_female:.efficacy_gmc_partner +
    .is_male:.efficacy_gmc_partner + (0 + .is_female + .is_male |
    couple_id)
Dispersion:                  ~0
Data: apim_distinguishable_data

     AIC       BIC    logLik -2*log(L)  df.resid
   605.4     625.9    -293.7     587.4        63

Random effects:

Conditional model:
 Groups    Name       Variance Std.Dev. Corr
 couple_id .is_female 388.3    19.70
           .is_male   286.0    16.91    0.79
Number of obs: 72, groups:  couple_id, 36

Conditional model:
                                 Estimate Std. Error z value Pr(>|z|)
.is_female                         32.853      3.342   9.829  < 2e-16 ***
```

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Verify model adequacy, distinguishability, and robustness

1. Did estimation succeed?
2. Does the model reproduce the important data patterns?
3. Is distinguishability supported?
4. Are conclusions robust or driven by few dyads or consequential modeling choices? (sensitivity analyses)

Did estimation succeed / converge?

The model converged without warnings… a good first sign!

Other things to look out for in the model output:

- Estimated variances near 0 or correlations near $\pm 1 \rightarrow$ possible boundary fit
- `NaN` or infinite estimates, standard errors, or model fit indices

Residual diagnostics / reproduction of data patterns

Residual diagnostics are not as straightforward anymore

- Linear models: straightforward checks
- Mixed models: residuals arise at multiple levels
- Generalized models: expectations depend on the family and link
- Dyadic models: partners' residuals are correlated

Solution: Compare the data with simulations from the fitted model.

DHARMa: one diagnostic scale across models

The `DHARMa` package (Hartig, 2026):

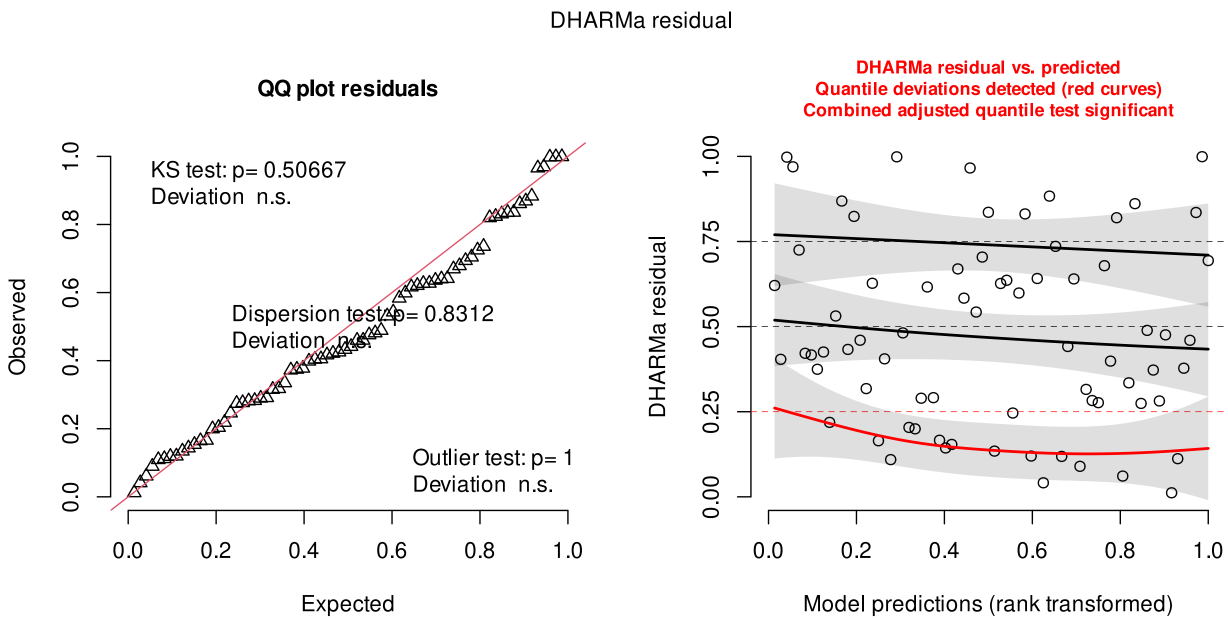
- Simulates outcomes from the fitted model
- Checks whether the observed residuals behave as expected under simulations from the fitted model.
- Adequate fit $\longrightarrow$ approximately uniform, pattern-free residuals

One function works across many model families!

Simulate the full dyadic residual structure

```r
library(DHARMa)

apim_distinguishable_dharma_residuals <- DHARMa::simulateResiduals(
  fittedModel = apim_distinguishable_model,
  n = 10000,

  # Since our model uses dispformula = ~ 0, we cannot use DHARMa's default
  # conditional simulations. With unconditional, the random-effects block representing
  # our residuals is re-simulated.
  simulateREs = "unconditional",

  # Male and female residuals are correlated. DHARMa estimates this correlation
  # from the simulations and rotates the residuals to "remove" it.
  rotation = 'estimated',
  seed = 123
)
```

Inspect DHARMa residuals



In this model:

- Uniformity and dispersion are plausible
- No simulation outliers
- A fitted-value pattern remains

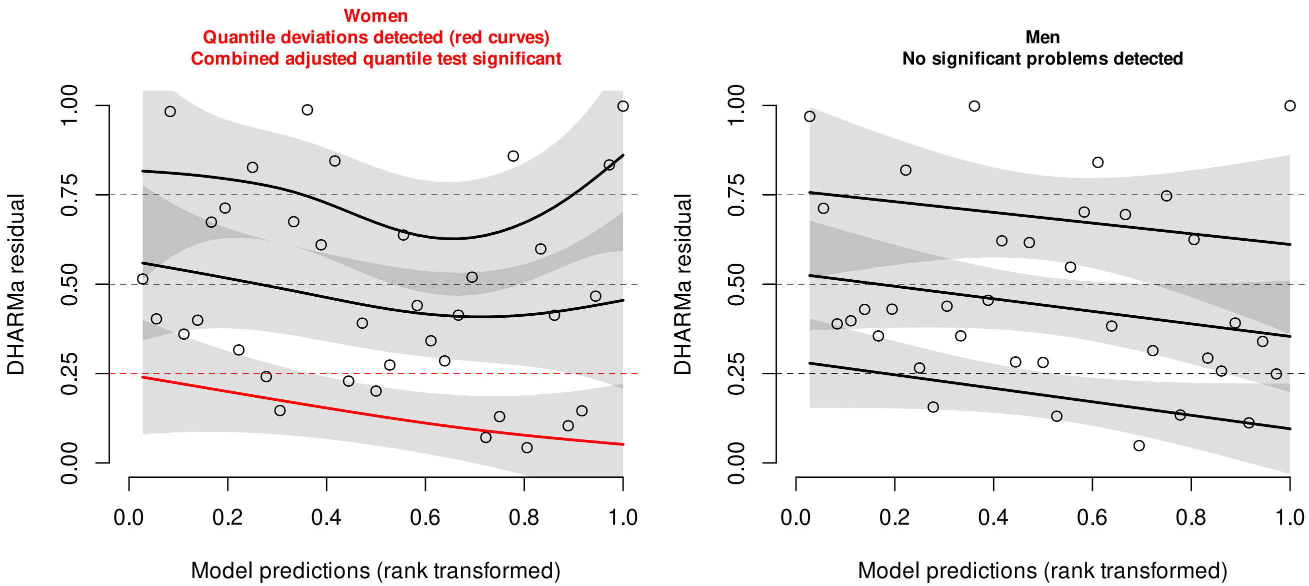
Next: inspect roles without rotation.

More targeted checks and guidance on interpretation of the plots: DHARMa vignette.

Inspect DHARMa residuals by role

```r
apim_distinguishable_female_dharma_residuals <- DHARMa::recalculateResiduals(
  apim_distinguishable_dharma_residuals,

  # Recalculate from the same simulations for one role.
  # One observation per couple: no rotation needed.
  sel = ~ .is_female == 1,
  rotation = NULL,
  seed = 123
)

apim_distinguishable_male_dharma_residuals <- DHARMa::recalculateResiduals(
  apim_distinguishable_dharma_residuals,
  sel = ~ .is_male == 1,
  rotation = NULL,
  seed = 123
)
```

The fitted-value pattern appears among women



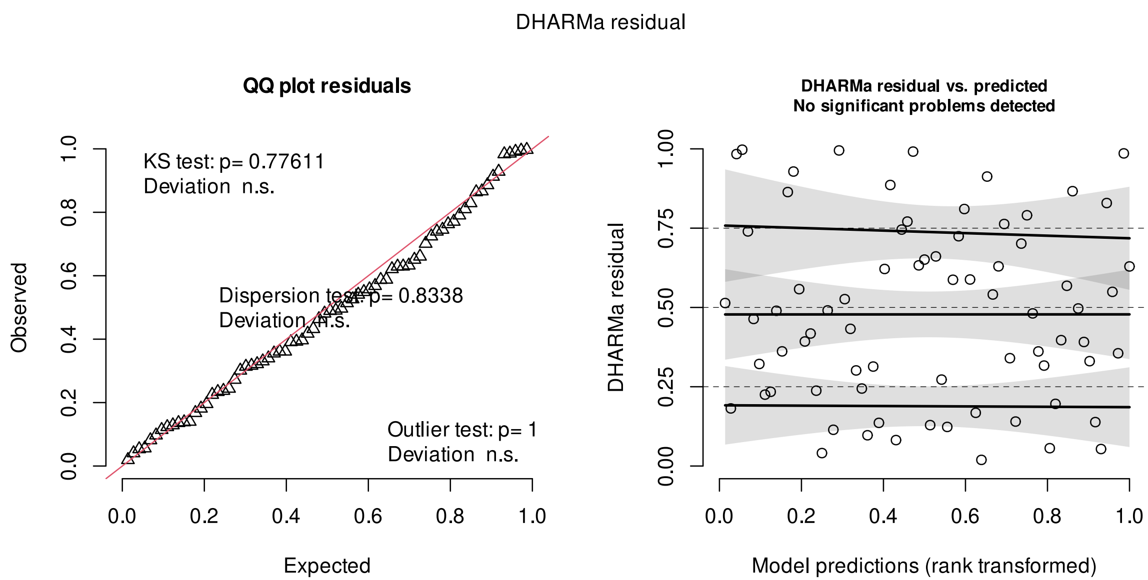
In this model:

- Pattern detected among women
- No comparable pattern among men

Next: try a defensible outcome scale.

More targeted checks and guidance on interpretation of the plots: DHARMa vignette.

Square-root MVPA reduces the fitted-value pattern



Compared with raw MVPA:

- Uniformity and dispersion are plausible
- No simulation outliers
- No pooled fitted-value pattern

Substantive result unchanged: actor effects remain positive and supported; partner effects remain unsupported.

In this example:

- **Sensitivity analysis:** square-root MVPA; conclusions unchanged
- **Primary analysis:** report the prespecified raw-MVPA model because the deviation was modest and minutes are directly interpretable
- **General rule:** larger or consequential diagnostic problems: revise main model

Transform—or change the outcome family?

- **Transformation may help** for a nonnegative, right-skewed outcome
  - Try `sqrt()` or `log1p()` (handles zeros)
  - Refit and rerun DHARMa; the histogram alone is not sufficient
- **A non-Gaussian family may be preferable**
  - Particularly for counts, proportions, bounded, or zero-heavy outcomes
  - Family and link choices require outcome-specific care
- **Workshop scope:** generalized models are not covered here

Test distinguishability

- Fit model with exchangeability constraints
- Compare models
- Pool only if theoretically defensible, aligned with RQ and theory, and compatible with the data
    - Depending on goals you can also report the distinguishable model and report results from this analysis

However, non-significant comparisons are not proof that roles are identical.

Gistelinck et al. (2018)

Estimate and compare:

```r
dyadMLM::compare_nested_models(
  apim_distinguishable_model, apim_exchangeable_comparison_model
)
```

```
Likelihood-ratio test for nested models fitted to equivalent data
Assumes mathematical nesting and an appropriate chi-squared reference distribution.

                                   Df    AIC    BIC  logLik deviance  Chisq
apim_exchangeable_comparison_model  5 604.25 615.64 -297.13   594.25
apim_distinguishable_model          9 605.44 625.93 -293.72   587.44 6.8155
                                   Chi Df Pr(>Chisq)
apim_exchangeable_comparison_model
apim_distinguishable_model              4      0.146

Conclusion (5% level): The likelihood-ratio test finds no clear improvement from
`apim_exchangeable_comparison_model` to `apim_distinguishable_model` (p = 0.146). Based on this test,
prefer `apim_exchangeable_comparison_model` for parsimony. This does not establish equal fit.
```

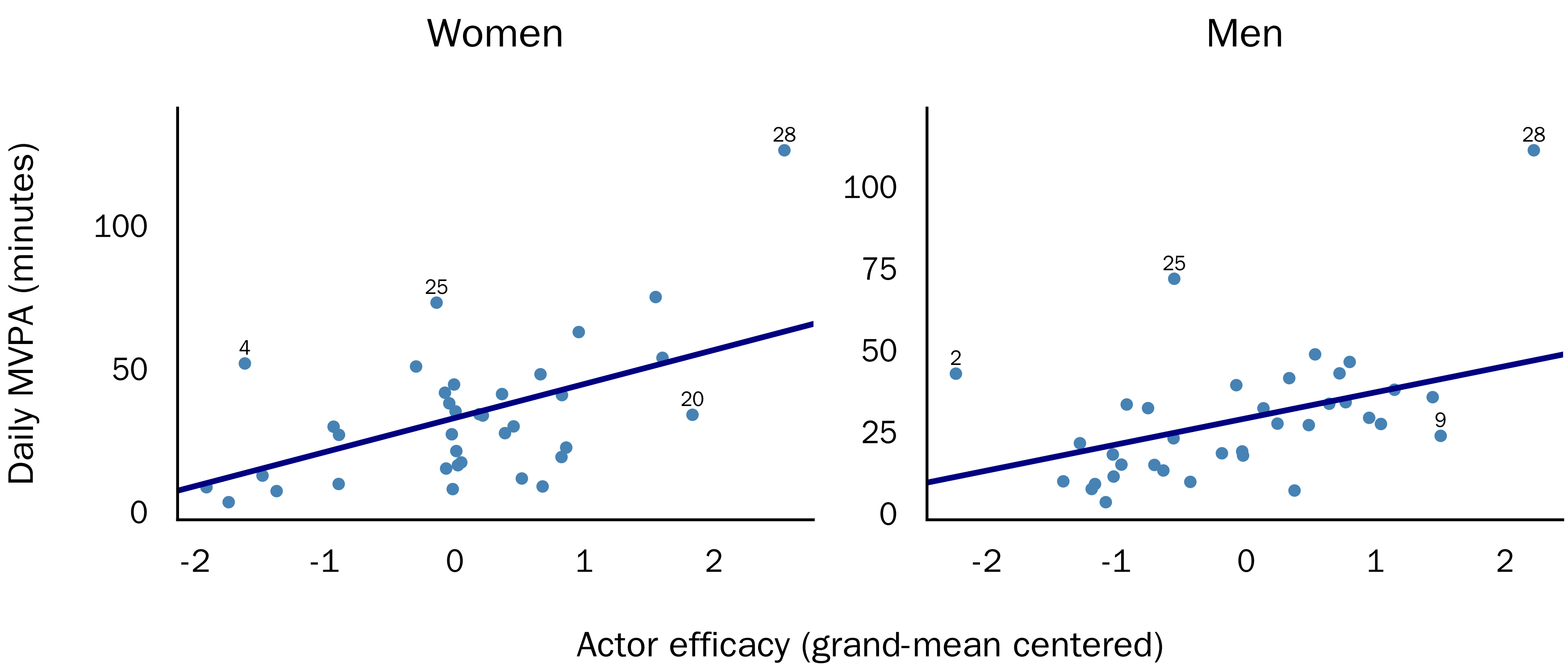
Test robustness - are conclusions driven by few dyads?

```r
plot_actor_association <- function(data, model, role, title,
                                   n_mahalanobis = 3) {
  xy <- cbind(data$.efficacy_gmc_actor, data$total_mvpa)
  labels <- data |>
    dplyr::mutate(.distance = mahalanobis(xy, colMeans(xy), cov(xy))) |>
    dplyr::slice_max(.distance, n = n_mahalanobis)
  coefficients <- glmmTMB::fixef(model)$cond

  ggplot(data, aes(.efficacy_gmc_actor, total_mvpa)) +
    geom_point(color = "steelblue") +
    geom_abline(intercept = coefficients[role],
                slope = coefficients[paste0(role, ":.efficacy_gmc_actor")],
                color = "navy", linewidth = 16 / 11) +
    geom_text(
      data = labels, aes(label = couple_id),
      vjust = -0.6, check_overlap = TRUE, size = 4
    ) +
    labs(
      title = title,
      x = "Actor efficacy (grand-mean centered)",
      y = "Daily MVPA (minutes)"
    ) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.12))) +
    see::theme_modern(base_size = 16) +
```

Test robustness - are conclusions driven by few dyads?

```r
n_mahalanobis <- 4

see::plots(
  plot_actor_association(
    dplyr::filter(apim_distinguishable_data, gender == "female"),
    apim_distinguishable_model, ".is_female", "Women",
    n_mahalanobis = n_mahalanobis
  ),
  plot_actor_association(
    dplyr::filter(apim_distinguishable_data, gender == "male"),
    apim_distinguishable_model, ".is_male", "Men",
    n_mahalanobis = n_mahalanobis
  )
) +
  patchwork::plot_layout(
    axes = "collect",
    axis_titles = "collect"
  )
```

Test robustness - are conclusions driven by few dyads?



Labels: four largest Mahalanobis distances in each plot.

Omitting Couple 28 changes the associations

```r
apim_without_couple_28 <- update(
  apim_distinguishable_model,
  data = filter(apim_distinguishable_data, couple_id != 28)
)

actor_sensitivity <- parameters::compare_parameters(
  `Full data` = apim_distinguishable_model,
  `Without Couple 28` = apim_without_couple_28,
  keep = "\\.efficacy_gmc_actor"
)
```

Actor slope	Full data	Without Couple 28
Women	11.87**	8.44*
Men	7.97 *	5.23

* $p < .05$; ** $p < .01$. Couple 28 was flagged in both visual screens. Its omission weakens both slopes; only the men's result crosses $p = .05$.

Other sensitivity analyses you could do

Fit a small set of (prespecified) models to test:

- coding decisions
- different plausible outcome families or transformations
- missing-data handling
- covariates with a substantive confounding or precision rationale

Workflow

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6. Visualize, interpret, and report results

Reporting results and answering the research question

```r
library(parameters)

parameters::model_parameters(apim_distinguishable_model, effects = 'fixed') |>
  insight::print_md(footer = "")
```

Fixed Effects

Parameter	Coefficient	SE	95% CI	z	p
is female	32.85	3.34	(26.30, 39.40)	9.83	< .001
is male	29.17	2.87	(23.55, 34.79)	10.17	< .001
is female × efficacy gmc actor	11.87	4.14	(3.75, 19.98)	2.87	0.004
is male × efficacy gmc actor	7.97	3.61	(0.89, 15.05)	2.21	0.027
is female × efficacy gmc partner	3.09	4.21	(-5.16, 11.34)	0.73	0.463
is male × efficacy gmc partner	4.05	3.55	(-2.91, 11.01)	1.14	0.254

In this sample, higher self-efficacy was associated with more of one's own MVPA for both women and men, but not clearly with the partner's MVPA.

Reporting results and answering the research question

Random effects in our model represent the residual structure

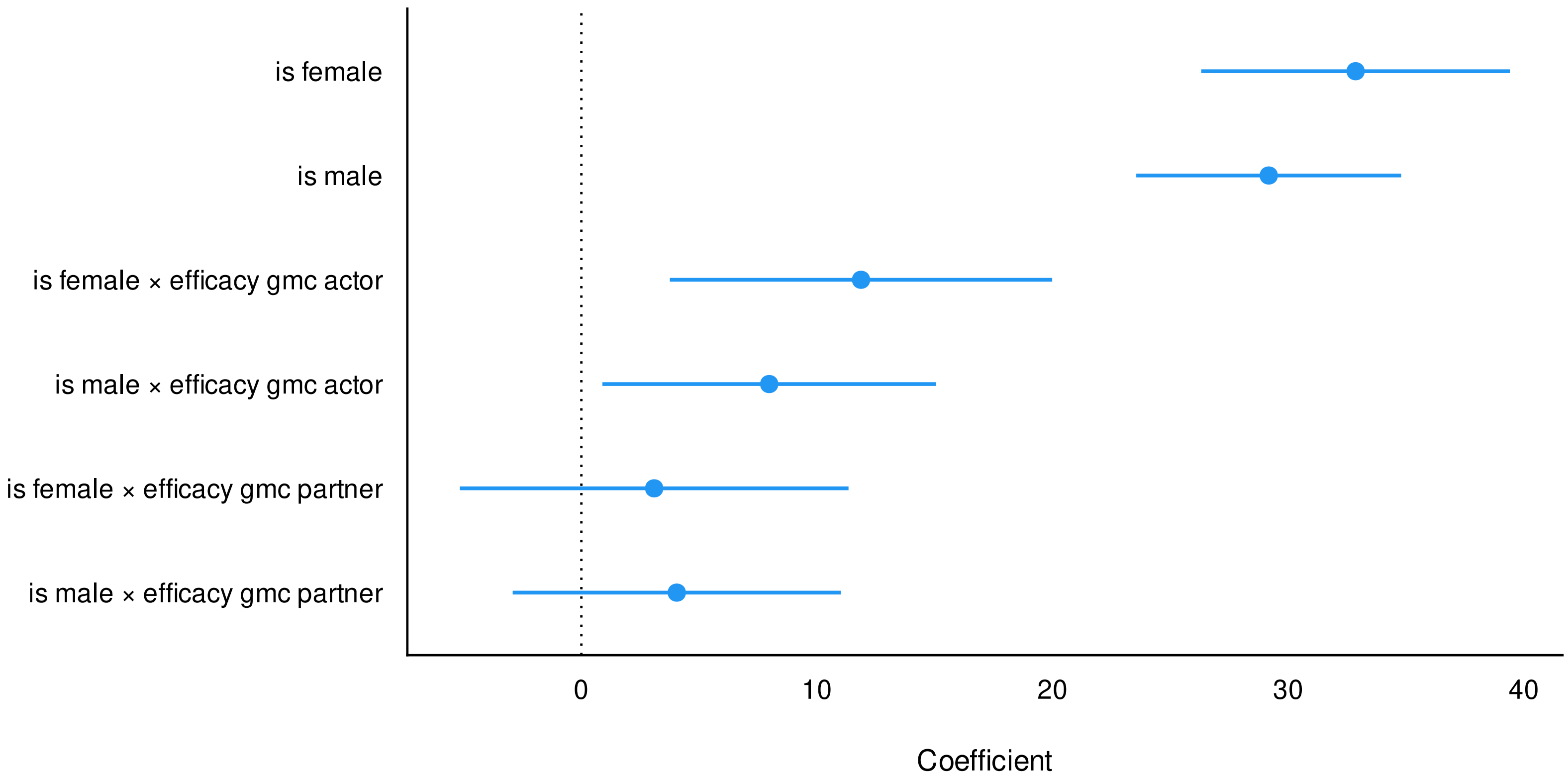
```
parameters::model_parameters(apim_distinguishable_model, effects = 'random') |>
  insight::print_md(footer = "")
```

Random Effects

Parameter	Coefficient	95% CI
SD (.is_female: couple_id)	19.70	(15.64, 24.83)
SD (.is_male: couple_id)	16.91	(13.42, 21.31)
Cor (.is_female~.is_male: couple_id)	0.79	

Visualize results

```
parameters::model_parameters(apim_distinguishable_model, effects = "fixed") |>
  plot()
```



What should be reported?

- **Analysis sample:** dyads and observations, exclusions, aggregation, and missing-data handling
- **Model specification:** outcome family and link, fixed effects, dyadic covariance structure, and equality constraints
- **Estimation and model decision:** estimator, software and version, convergence, boundary estimates, distinguishability comparison, and retained model
- **Results:** focal estimates with 95% confidence intervals in substantive units and the fitted variance-covariance parameters
- **Checks and interpretation:** residual diagnostics, influential dyads, prespecified sensitivity analyses, material changes, and causal limits

Exchangeable dyads

The same model but constrained

Workflow

1. Define dyads, distinguishability, and research questions
2. Prepare the data and validate the dyadic structure
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Define dyads, distinguishability, and research questions

> Same data and RQ, now with exchangeability constraints for illustration.

Workflow

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Create the required format and validate the data

You can use `dyadMLM::prepare_dyad_data()` to prepare and validate the data.

For exchangeable dyads, we can just omit the `role` argument. If we want to estimate both distinguishable and exchangeable versions, we would pass `role` and set `include_arbitrary_member_contrast = TRUE`.

```
apim_exchangeable_data <- dyadMLM::prepare_dyad_data(
  data = raw_dyad_data,
  dyad = couple_id,
  member = person_id,
  predictors = efficacy,
  model_types = 'apim',
  add_apim_gmc_predictors = TRUE,
  seed = 123
)

print(apim_exchangeable_data)
```

```
# dyadMLM data
# Rows: 72 | Dyads: 36 | Intensive longitudinal: no
# Structure: dyad = couple_id, member = person_id
#
# Dyad compositions:
# assumed_exchangeable exchangeable 36 dyads
```

Create the required format and validate the data

Looking at the relevant columns:

```
head(apim_exchangeable_data) |>
  dplyr::select(
    couple_id, person_id, .member_contrast_arbitrary, .efficacy_gmc, .efficacy_gmc_actor, .efficacy_g
  ) |>
  slide_table()
```

couple_id	person_id	.member_contrast_arbitrary	.efficacy_gmc	.efficacy_gmc_actor	.efficacy_gmc_partner	total_
1.00	1.00	-1.00	0.64	0.64	-1.49	
1.00	2.00	1.00	-1.49	-1.49	0.64	
2.00	3.00	-1.00	-2.24	-2.24	-0.01	
2.00	4.00	1.00	-0.01	-0.01	-2.24	
3.00	5.00	-1.00	0.96	0.96	-0.76	
3.00	6.00	1.00	-0.76	-0.76	0.96	

Workflow

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Describe focal variables (pooled)

No wide-view of the data is needed for exchangeable dyads. We can describe the data directly to obtain pooled estimates:

```
apim_exchangeable_data |>
  dplyr::select(efficacy, total_mvpa) |>
  report::report_table() |>
  slide_table()
```

Variable	n_Obs	Mean	SD	Median	MAD	Min	Max	Skewness	Kurtosis	n_Missing
efficacy	72.00	2.38	0.99	2.38	1.06	0.13	4.92	0.09	-0.15	0.00
total_mvpa	72.00	31.21	22.40	27.55	18.67	3.43	126.35	1.89	5.34	0.00

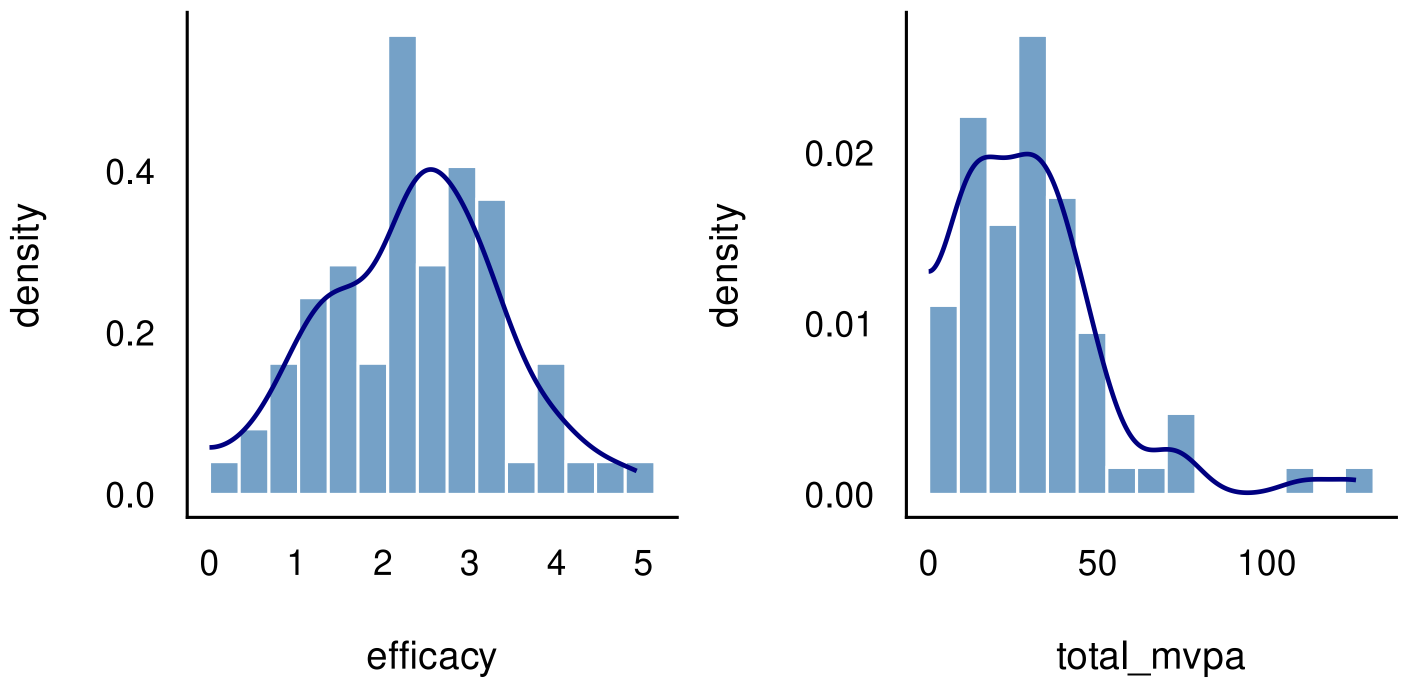
Visualize focal variables

```r
plot_pooled_histogram <- function(data, var) {
  var_label <- rlang::as_label(rlang::enquo(var))

  ggplot(
    data,
    aes(x = {{ var }}, y = after_stat(density))
  ) +
    geom_histogram(
      bins = 15, boundary = 0,
      fill = "steelblue", color = "white", alpha = 0.75
    ) +
    geom_density(
      bounds = c(0, Inf), color = "navy", linewidth = 1.1
    ) +
    see::theme_modern(base_size = 16)
}
```

Visualize focal variables

```r
see::plots(
  plot_pooled_histogram(apim_exchangeable_data, efficacy),
  plot_pooled_histogram(apim_exchangeable_data, total_mvpa)
)
```



Observed correlations

For exchangeable dyads, ICCs summarize partner similarity:

```
library(wbCorr)

apim_exchangeable_correlations <- wbCorr(
  apim_exchangeable_data[,c('efficacy','total_mvpa')],
  cluster = apim_exchangeable_data$couple_id
)

summary(apim_exchangeable_correlations, 'wb')
```

```
$merged_wb
           efficacy total_mvpa
efficacy     [0.60]     0.44**
total_mvpa  0.56***     [0.79]

$note
[1] "***p < 0.001, **p < 0.01, *p < 0.05"
```

Diagonal ICCs: expected correlation between partners on each variable

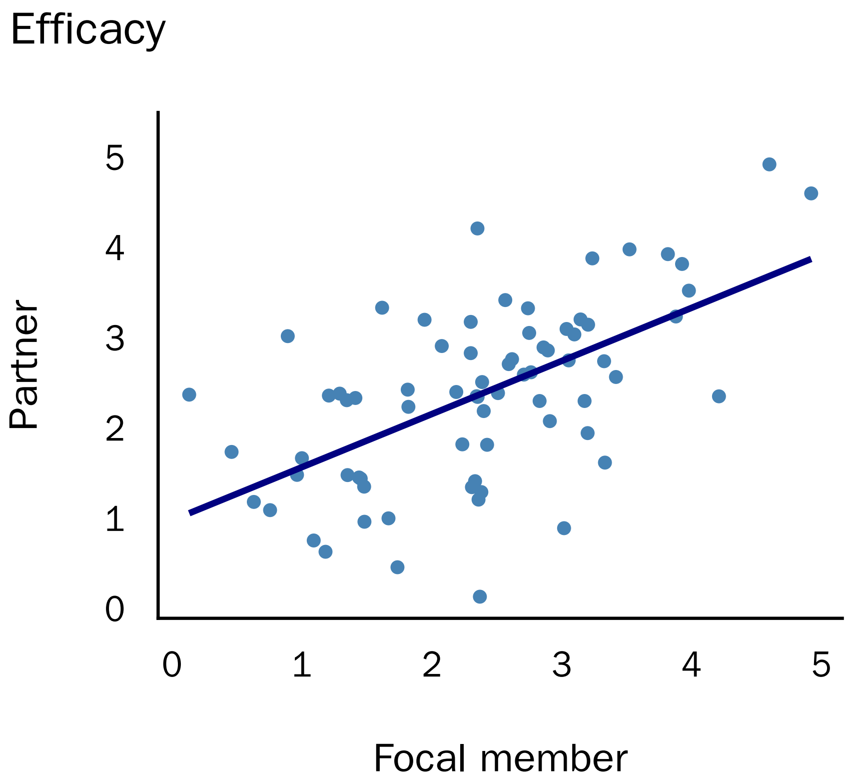
Off-diagonals: top right = within-couple correlation; bottom left = between-couple correlation

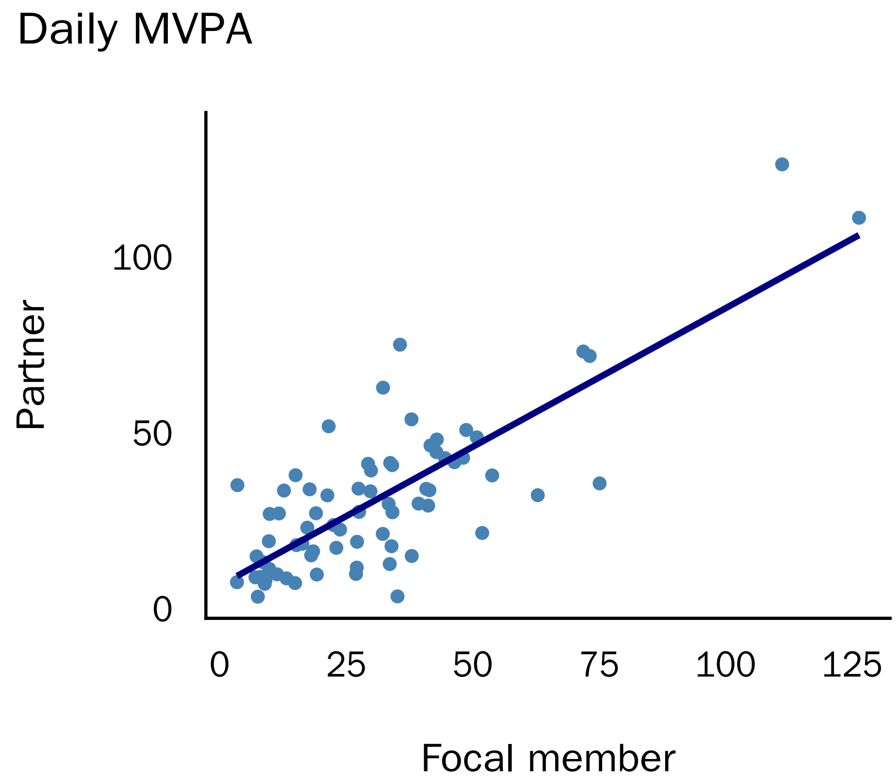
Visualize partner similarity

```r
partner_similarity_data <- apim_exchangeable_data |>
  dplyr::group_by(couple_id) |>
  dplyr::mutate(
    efficacy_partner = rev(efficacy),
    total_mvpa_partner = rev(total_mvpa)
  ) |>
  dplyr::ungroup()

see::plots(
  plot_partner_similarity(
    partner_similarity_data,
    efficacy, efficacy_partner, couple_id, "Efficacy",
    n_mahalanobis = 0
  ) +
    labs(x = "Focal member", y = "Partner"),
  plot_partner_similarity(
    partner_similarity_data,
    total_mvpa, total_mvpa_partner, couple_id, "Daily MVPA",
    n_mahalanobis = 0
  ) +
    labs(x = "Focal member", y = "Partner")
)
```

Visualize partner similarity





Each couple is shown twice—once with each person as the focal member.

Workflow

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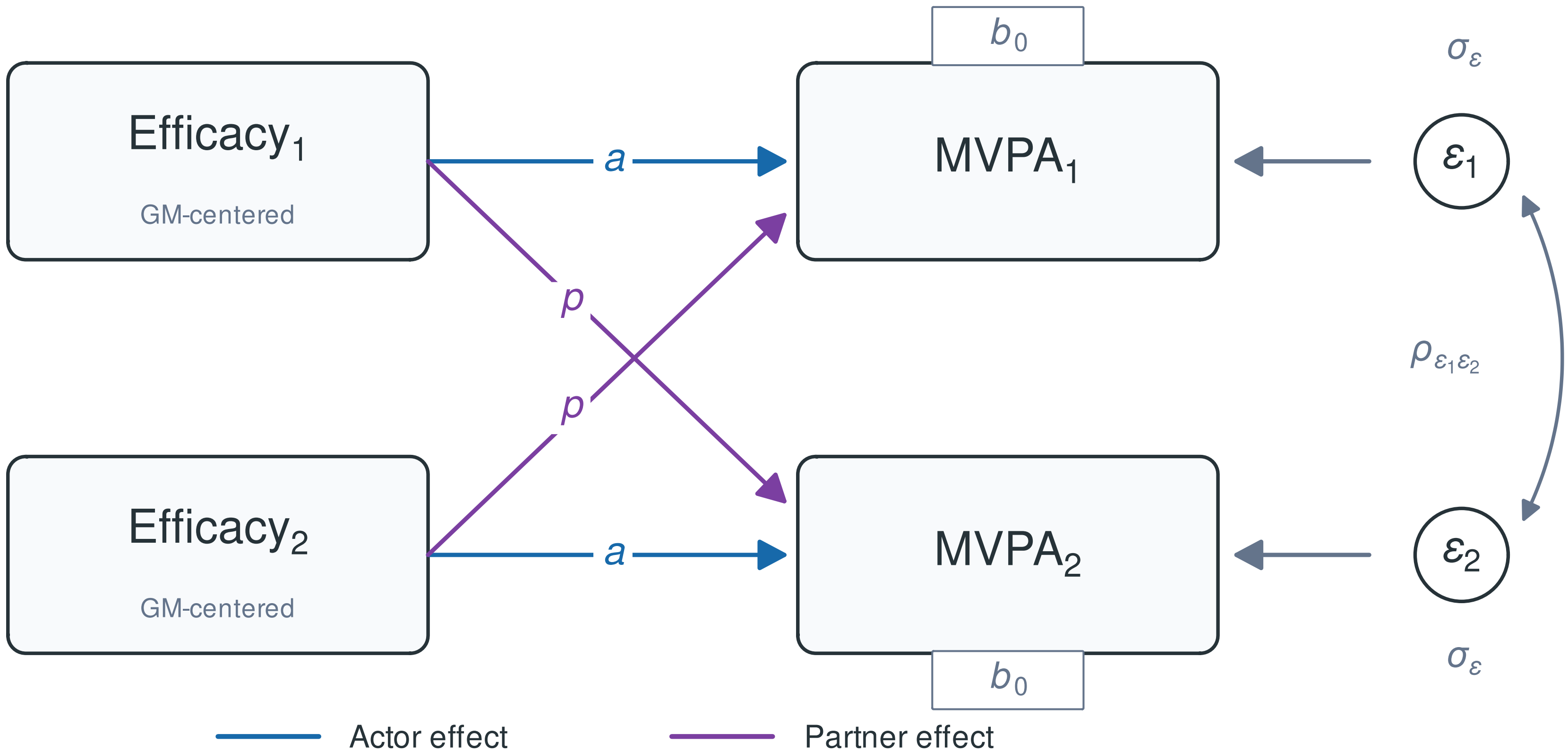
Exchangeable APIM



In MLM we constrain / pool by omitting terms

Both partners' formulas are identical, so no need to combine them. We can just write:

For each dyad i, each member j has their own predictor $X_{j,i}^{(\mathrm{A})}$ and their partner's predictor $X_{j,i}^{(\mathrm{P})}$:

$$Y_{j,i} = b_0 + a \times X_{j,i}^{(\mathrm{A})} + p \times X_{j,i}^{(\mathrm{P})} + \epsilon_{j,i}$$

- Actor effects - Partner effects

In long format, each equation applies to one member row of the dyad.

The model pools the effects, but actor and partner values are exchanged between member rows, so their predictions may still differ.

The tricky part: residuals need to remain correlated within dyads

- The two member rows from the same dyad retain a joint residual structure
- Both members now have the same residual variance (pooled)
- Position 1 and 2 within the dyad are arbitrary

$$\text{Cov}\begin{pmatrix} \epsilon_{1i} \\ \epsilon_{2i} \end{pmatrix} = \mathbf{\Sigma}_\epsilon = \begin{bmatrix} \sigma_\epsilon^2 & \rho_{\epsilon_1 \epsilon_2}\sigma_\epsilon^2 \\ \rho_{\epsilon_1 \epsilon_2}\sigma_\epsilon^2 & \sigma_\epsilon^2 \end{bmatrix}$$

Fitting the exchangeable APIM in glmmTMB

```
apim_exchangeable_model <- glmmTMB(
  total_mvpa ~
    # Pooled intercept ------------------------------------
    1 +

    # Pooled actor effect ----------------------------------
    .efficacy_gmc_actor +

    # Pooled partner effect --------------------------------
    .efficacy_gmc_partner +

    # Within-dyad residual covariance (Sum/Deviation) ------

    # Residuals of partners might be similar (positively correlated)
    (1 | couple_id) +

    # Residuals of partners might move opposite (negatively correlated)
    (0 + .member_contrast_arbitrary | couple_id),

  dispformula = ~ 0,
  family = gaussian(),
  data = apim_exchangeable_data
)
```

The mean-deviation parameterization

- The DIM represents member residuals using two uncorrelated components:
  - Mean block `(1 | couple_id)`: $r_{\mathrm{M}i} = (\epsilon_{1i} + \epsilon_{2i})/2$; both members move together
  - Deviation block `(0 + .member_contrast_arbitrary | couple_id)`: $r_{\mathrm{D}i} = (\epsilon_{1i} - \epsilon_{2i})/2$; members move in opposite directions

Workflow

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Verify model adequacy, distinguishability, and robustness

Essentially the same steps and code can be used to check:

1. Did estimation succeed?
2. Does the model reproduce the important data patterns?
3. Does a distinguishable model fit clearly better?
4. Are conclusions robust or driven by few dyads or consequential modeling choices? (sensitivity analyses)

Workflow

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Reporting results and answering the research questions

```r
parameters::model_parameters(apim_exchangeable_model, effects = 'fixed') |>
  insight::print_md(footer = "")
```

Fixed Effects

Parameter	Coefficient	SE	95% CI	z	p
(Intercept)	31.21	2.89	(25.54, 36.88)	10.79	< .001
efficacy gmc actor	10.34	2.04	(6.35, 14.34)	5.07	< .001
efficacy gmc partner	3.17	2.04	(-0.83, 7.17)	1.55	0.120

In the parsimonious exchangeable model, higher self-efficacy was associated with more of one's own MVPA, but not clearly with the partner's MVPA.

Reporting results and answering the research questions

```r
parameters::model_parameters(apim_exchangeable_model, effects = 'random') |>
  insight::print_md(footer = "")
```

Random Effects

Parameter	Coefficient	95% CI
SD (Intercept: couple_id)	17.36	(13.78, 21.87)
SD (.member_contrast_arbitrary: couple_id)	6.48	

Often it is preferable to rotate the covariance back to the familiar member-level covariance matrix:

- Member residual variance: $\mathrm{Var}(\epsilon_{1i}) = \mathrm{Var}(\epsilon_{2i}) = \mathrm{Var}(r_{\mathrm{M}i}) + \mathrm{Var}(r_{\mathrm{D}i})$
- Partner residual covariance: $\mathrm{Cov}(\epsilon_{1i}, \epsilon_{2i}) = \mathrm{Var}(r_{\mathrm{M}i}) - \mathrm{Var}(r_{\mathrm{D}i})$

Reporting results and answering the research questions

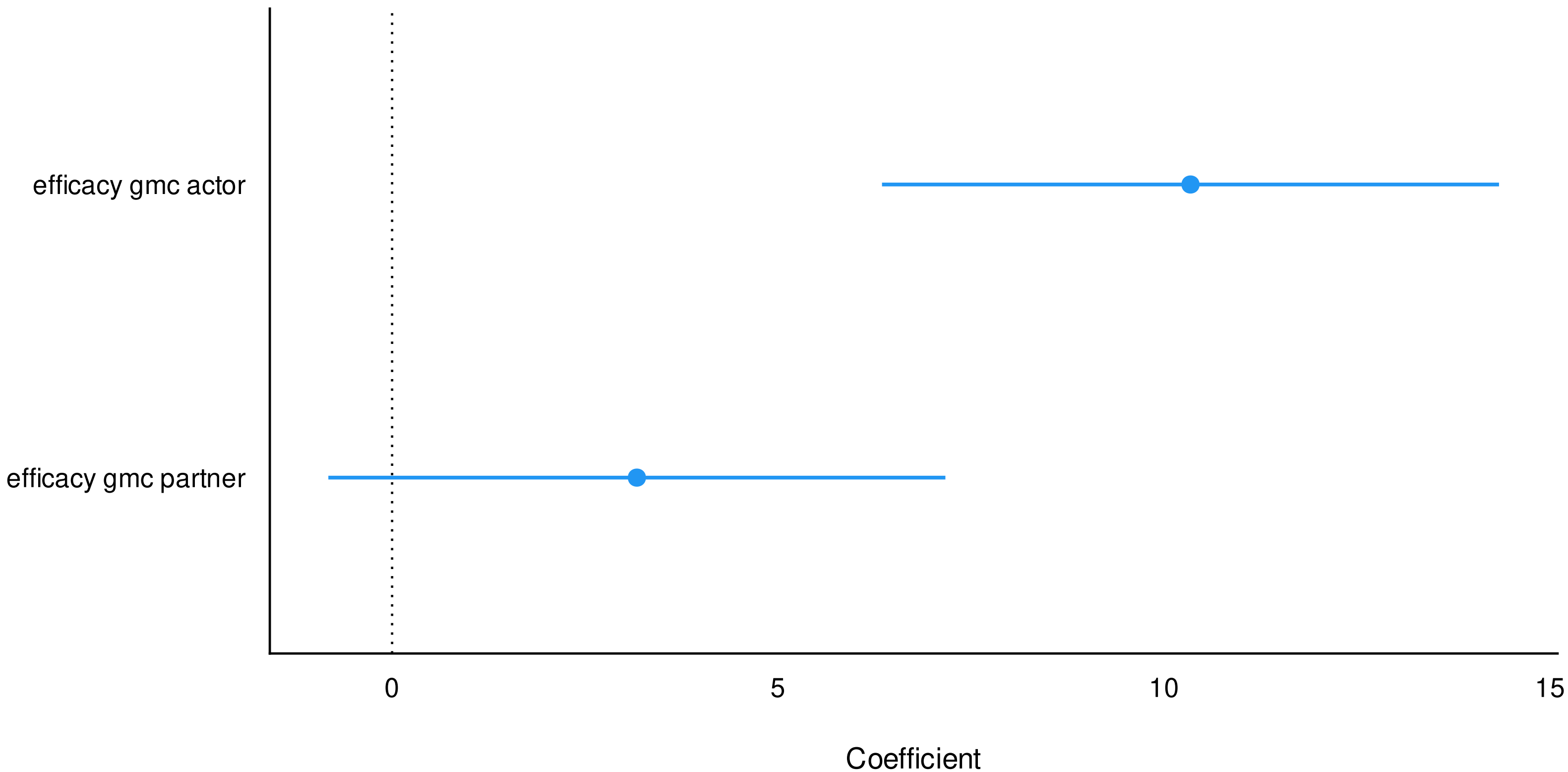
No need to do so by hand. Rotating back with `dyadMLM`:

```r
dyadMLM::recover_exchangeable_covariance(apim_exchangeable_model)
```

```
Exchangeable residual covariance

Pair `pair_1`
Shared:     us(1 | couple_id)
Difference: us(0 + .member_contrast_arbitrary | couple_id)

Variance-covariance:
                        1       2
1 member1: (Intercept) 343.216 259.269
2 member2: (Intercept) 259.269 343.216

Standard deviations and correlations:
                        1      2
1 member1: (Intercept) 18.526 0.755
2 member2: (Intercept) 0.755  18.526
```

Visualize results

```r
parameters::model_parameters(apim_exchangeable_model, effects = 'fixed') |>
  plot()
```



Optional: DIM and DSM

Worked examples are available in the package vignettes of `dyadMLM`.

Open the DIM vignette →

Open the DSM vignette →

Intensive longitudinal dyads

ILD workflow

The same core workflow

1. Define dyads, distinguishability, and research questions
2. Prepare the data and validate the dyadic structure
3. Describe and visualize measures and observed relationships
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ILD additionally requires

- Check compliance, matched occasions, and actual time gaps (Revol et al., 2024)
- Plot paired trajectories and inspect raw temporal and cross-partner patterns (Bolger & Laurenceau, 2013; Revol et al., 2024)
- Separate within- and between-person variation
  - ICCs, level-specific correlations, and multilevel reliability where applicable (Bolger & Laurenceau, 2013; Geldhof et al., 2014; Hamaker, 2024)
- Choose the temporal specification
  - Time scale and lag, trends/stationarity, concurrent versus lagged effects, and discrete versus continuous time (Driver et al., 2017)
- Model dyads on multiple levels
- Check residual temporal and cross-partner dependence

Synthesized from Bolger & Laurenceau (2013), Revol et al. (2024), McNeish & Hamaker (2020), and del Rosario & West (2025).

ILD Workflow

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The Dataset

Same as before:

- 36 physically inactive romantic couples
- Both partners intended to become more physically active
- Smartphone-based planning interventions and JITAIs
- Daily diaries over 55 days

Not aggregated anymore:

- Daily total MVPA in minutes
- Daily self-efficacy (0–5): “I am sure I can be active tomorrow”
- Self-efficacy on day $t - 1 \rightarrow$ physical activity on day t

ILD Research Question

RQ: **How are daily deviations and stable between-person differences in self-efficacy associated with one's own and one's partner's physical activity the next day?**

Distinguishable ILD APIM

Distinguishing by gender

ILD Workflow

1. Define dyads, distinguishability, and research questions
2. Prepare the data and validate the dyadic structure
3. Describe and visualize measures and observed relationships
4. Estimate the models
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6. Visualize, interpret, and report results

Load and inspect the raw data

```
raw_ild_dyad_data <- readRDS("dyadic-data.rds") |>
  dplyr::select(
    couple_id, person_id, diaryday, gender, efficacy, total_mvpa
  ) |> dplyr::arrange(couple_id, diaryday)

slide_table(head(raw_ild_dyad_data))
```

couple_id	person_id	diaryday	gender	efficacy	total_mvpa
1.00	1.00	0.00	male	5.00	60.00
1.00	2.00	0.00	female	2.00	30.00
1.00	1.00	1.00	male	5.00	60.00
1.00	2.00	1.00	female	2.00	90.00
1.00	1.00	2.00	male	5.00	60.00
1.00	2.00	2.00	female	2.00	0.00

Create the required format and validate the data

As before

- Actor and partner variables
- Arbitrary member contrast
- Dyad validation

For ILD

Index time by the outcome day t. For self-efficacy $X_{ri,t-1}$ measured on the previous day:

- Between-person component: Person mean centered on the mean of person means
  - $\bar{X}_{ri} - \bar{X}$
- Lagged within-person component: Previous-day deviation from the person mean
  - $X_{ri,t-1} - \bar{X}_{ri}$

Here, $\bar{X}_{ri}$ is member r's observed mean in dyad i, and $\bar{X}$ is the equally weighted mean of the observed person means.

Create the required format and validate the data

Manual code →

```r
ild_apim_data <- dyadMLM::prepare_dyad_data(
  data = raw_ild_dyad_data,
  dyad = couple_id,
  member = person_id,
  role = gender,
  time = diaryday,
  predictors = efficacy,
  lag1_predictors = efficacy,
  model_types = "apim",
  include_arbitrary_member_contrast = TRUE,
  seed = 123
)
```

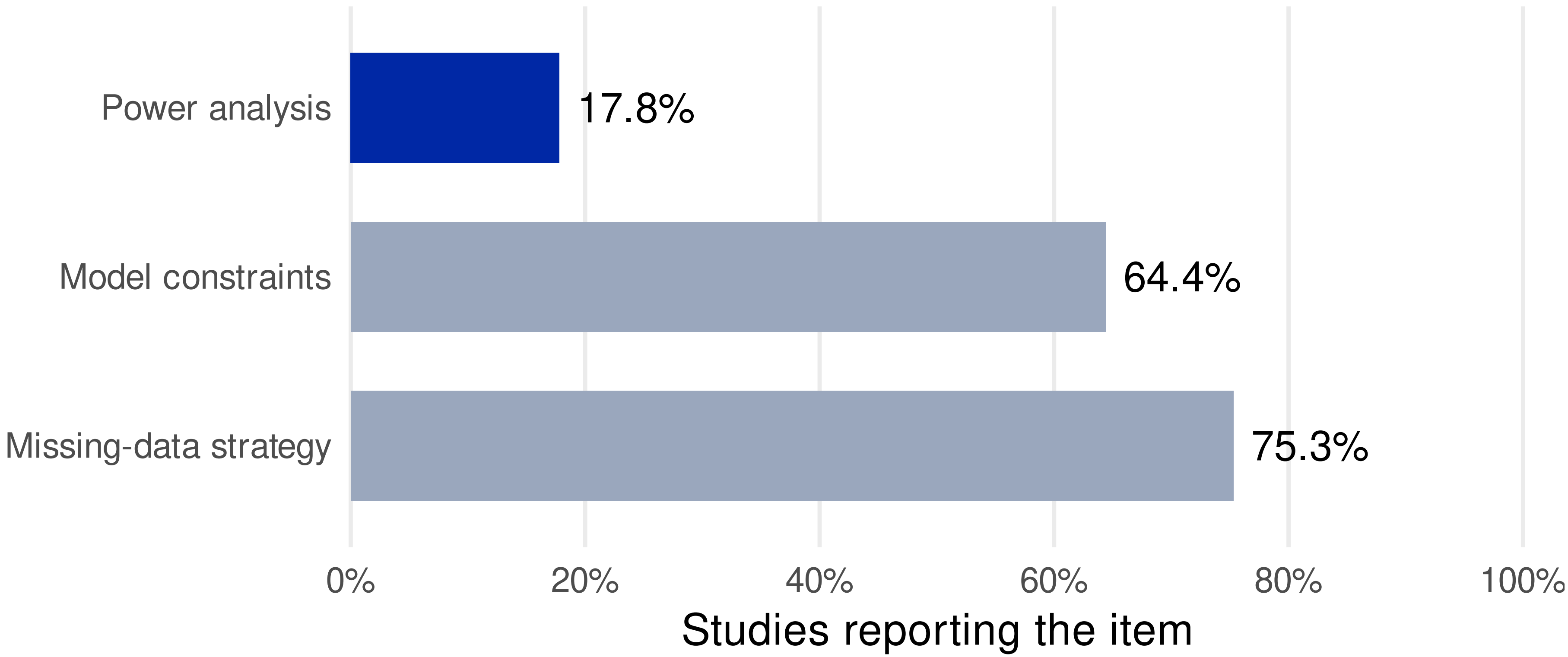
- Lagged CWP terms: day-$t-1$ efficacy $\rightarrow$ day-t MVPA
- CBP terms: stable between-person differences; not lagged
- Exact day matching: missing occasions are not bridged

Exchangeable APIM

AR(1), OU, and Dynamic Models

- When to use dynamic models vs. concurrent models
- Current possibilities in R MLM for AR(1) or OU models
- Current possibilities in R MLM for dynamic models
- Alternatives (DSEM in Mplus or lavaan for X)

Power analysis is the clearest reporting gap



Systematic review of 73 longitudinal APIM studies published from 2008–2025 (Mohammadzadeh Yazd et al., 2026).

Make every dyadic analysis decision visible

1. **Preregister** actor and partner hypotheses and key analysis decisions.
2. **Plan and report power:** use simulation matched to the complete planned model.
3. **Explain missing data:** describe missingness and the handling strategy.
4. **Specify the full model:** roles and distinguishability, actor and partner paths, constraints, and covariance structure.
5. **Report reliability where applicable:** within- and between-level omega for multilevel item scales, e.g., `multilevelTools::omegaSEM()` (Geldhof et al., 2014; Wiley, 2025).

The review also recommends multi-method measurement and more diverse populations

Mohammadzadeh Yazd et al. (2026)

Exercises in R

- Go to https://pascal-kueng.github.io/dyadMLM/workshop/
- Link should be in the Teams channel
- Download and unzip all workshop materials

Run this once in a fresh R session before the break. It installs or updates the workshop packages and any dependencies that need updating; this may take several minutes.

```r
source("00_setup.R")
```

References

Amireault, S., & Godin, G. (2015). The godin–shephard leisure-time physical activity questionnaire: Validity evidence supporting its use for classifying healthy adults into active and insufficiently active categories. *Perceptual and Motor Skills*, *120*(2), 604–622. https://doi.org/10.2466/03.27.PMS.120v19x7

Asparouhov, T., Hamaker, E. L., & Muthén, B. (2018). Dynamic structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(3), 359–388. https://doi.org/10.1080/10705511.2017.1406803

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Appendix

Create the cross-sectional APIM variables manually

← Data preparation

- Explicit matching by couple and role
- Missing partner ⟶ `NA`

```r
apim_manual_data <- raw_dyad_data |>
  dplyr::mutate(
    .is_female = as.numeric(gender == "female"),
    .is_male = as.numeric(gender == "male"),
    .efficacy_gmc = efficacy - mean(efficacy, na.rm = TRUE),
    .efficacy_gmc_actor = .efficacy_gmc
  )

partner_values <- apim_manual_data |>
  dplyr::transmute(
    couple_id,
    gender = dplyr::recode(gender, female = "male", male = "female"),
    .efficacy_gmc_partner = .efficacy_gmc
  )

apim_manual_data <- apim_manual_data |>
  dplyr::left_join(
    partner_values,
    by = c("couple_id", "gender"),
    relationship = "one-to-one"
  )
```

Decompose self-efficacy manually

← Data preparation

```
efficacy_components <- raw_ild_dyad_data |>
  dplyr::group_by(person_id) |>
  dplyr::mutate(
    .person_mean = mean(efficacy, na.rm = TRUE),
    .cwp = efficacy - .person_mean
  ) |>
  dplyr::ungroup() |>
  dplyr::mutate(
    .cbp = .person_mean -
      mean(.person_mean[!duplicated(person_id)], na.rm = TRUE)
  )
```

Match actor and partner components manually

← Data preparation

```
between_by_role <- efficacy_components |>
  dplyr::distinct(couple_id, gender, .cbp) |>
  tidyr::pivot_wider(names_from = gender, values_from = .cbp,
                     names_glue = ".cbp_{gender}")

lagged_within_by_role <- efficacy_components |>
  dplyr::transmute(couple_id, diaryday = diaryday + 1L, gender, .cwp) |>
  tidyr::pivot_wider(names_from = gender, values_from = .cwp,
                     names_glue = ".cwp_{gender}")

ild_manual_data <- raw_ild_dyad_data |>
  dplyr::left_join(between_by_role, by = "couple_id") |>
  dplyr::left_join(
    lagged_within_by_role,
    by = c("couple_id", "diaryday")
  ) |>
  dplyr::mutate(
    .efficacy_cbp_actor = dplyr::if_else(
      gender == "female", .cbp_female, .cbp_male
    ),
    .efficacy_cbp_partner = dplyr::if_else(
      gender == "female", .cbp_male, .cbp_female
    ),
    .efficacy_cwp_actor_lag1 = dplyr::if_else(
      gender == "female", .cwp_female, .cwp_male
    ),
    .efficacy_cwp_partner_lag1 = dplyr::if_else(
      gender == "female", .cwp_male, .cwp_female
    )
  )
```

- Explicit day-role matching: independent of row order; unmatched lags remain `NA`